

SafePassage AI

School & College Cab Management System

Complete System Architecture, Role-Based Operational Move Manual & Technical Master Brochure
A Unified Multi-Tenant Platform for Institutions, Fleets, Drivers, Students, Parents & Corporate Shuttles

■■ Strict Multi-Role Gateway Direct Super Admin approval (■ 30-min SLA or ■ Tomorrow batch).	■ 45-Second AI Dispatch Dynamic radar radius clustering and rapid trip matching algorithm.
■ Live Radar Telemetry Real-time student & vehicle tracking with geofence proximity alerts.	■ Automated Clearinghouse Razorpay fee escrow, automated payouts, and subscription ledgers.
■ 3 Native Mobile APKs React Native Parent App, Driver Cockpit & Admin Mobile Consoles.	■■ 6 Microservice Engines Spring Boot 3.2 distributed mesh on MySQL transactional database.

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Executive Summary

SafePassage AI is an enterprise-grade, holistic transportation intelligence platform designed to eliminate the logistical chaos, security risks, and administrative overhead associated with educational institution cabs, commercial fleet networks, college student transit, and corporate employee shuttles. By combining high-frequency GPS telemetry, automated 45-second dispatch heuristics, dynamic stop clustering, and a rigorous multi-role authorization framework, SafePassage AI provides real-time transparency and safety to all stakeholders.

The system operates across six distinct user personas with customized user interfaces (both responsive web prototypes and native React Native mobile applications), powered by a decoupled Spring Boot microservices backend and an enterprise MySQL relational persistence tier.

CURRENT DEPLOYMENT STATUS: ACTIVE & VERIFIED

System Milestone Summary: All 6 microservices (8080–8085), the web prototype (5173), and the 3 Android React Native mobile apps (Parent 8092, Driver 8091, Admin 8093) have been fully built, verified, and linked to automated 1-click desktop launchers with robust Intel UHD hardware acceleration.

Section 1: Executive Overview & Unified Ecosystem Architecture

1.1 The Core Problem & Value Proposition

Conventional institutional transport relies on manual paper rosters, unverified drivers, unreliable phone calls from anxious parents, and unmonitored routes. SafePassage AI solves these pain points through four foundational pillars:

- **Zero-Trust Safety & Verification:** Strict Super Admin approval protocol for every registered driver, vehicle, and student pass before authorization.
- **Live Telemetry & Radar Proximity:** High-precision GPS streaming with automated 500m and 100m geofence alerts pushed to parents and riders.
- **45-Second AI Dispatch Protocol:** Algorithmic matching for ad-hoc and single-trip transport requests with boarding PIN security.
- **Unified Clearinghouse:** Automated Razorpay subscription collections, driver payouts, and institutional billing reconciliation.

1.2 Ecosystem Personas & Matrix

Role Persona	Primary Interface	Core Responsibilities	SLA / Security Policy
Super Admin	Web Console (5173) Admin Mobile APK	Master governance, multi-institution onboarding, direct registration approvals, financial settlements, emergency broadcast.	Master Password protected; Instant or Scheduled activation authority.
Cab Driver	Driver Mobile APK (8091) Web Cockpit	Shift toggle, turn-by-turn waypoint navigation, student boarding roster, 45s trip acceptance, SOS beacon.	License validation; Strict approval required; 4-digit passenger OTP boarding.
Cab Owner	Fleet Web Portal (5173)	Multi-vehicle management, driver shift assignment, fuel/maintenance logs, contract bidding, revenue analytics.	Commercial vehicle permit, RC book, and fitness certificate verification.
Parent	Parent Mobile APK (8092) Web Portal	Live bus radar tracking, student pickup/drop alerts, digital fee payment, driver emergency call, absence marking.	Student linked account; Proximity push alerts at 500m & 100m.
Student	Student Web Hub (5173) Pass APK	Digital campus pass, daily shuttle live locator, peer ride-share booking, split-fare calculation, SOS emergency link.	Institution roll number & semester identity validation.
Working Pro	Corporate Web Shuttle (5173)	IT Park / Corporate daily shuttle subscription, desk-to-door dynamic routing, expense voucher reimbursement.	Corporate employee ID & office shift validation.

1.3 High-Level Solution Topology

The SafePassage AI ecosystem consists of 3 synchronized tiers: the Client Tier (Vite Web Prototype & React Native Mobile Apps), the Microservices Gateway Mesh (6 Spring Boot services), and the Storage & Clearinghouse Tier (MySQL Relational Database & Razorpay Webhooks). All components communicate via low-latency REST APIs and WebSocket telemetry endpoints.

Section 2: Authentication Gateway & Direct Super Admin Approval

2.1 Direct Super Admin Approval vs. Forced OTP Workflow

To provide a frictionless yet highly secure onboarding experience, SafePassage AI has transitioned to a **Direct Super Admin Acceptance** model. Users are **not** burdened with entering complex 6-digit OTP codes for new registrations. Instead, when a new driver, owner, parent, student, or professional submits their registration, their application is dispatched directly into the Super Admin's Central Authorization Desk.

DIRECT REGISTRATION APPROVAL PROTOCOL

Two Flexible Approval SLAs for Registrations:

- 1. **Accept Now (within 30 Minutes):** Fast-track emergency review for immediate driver shifts or urgent student cab enrollments.
- 2. **Accept for Tomorrow:** Standard batch processing for upcoming semester admissions or scheduled fleet expansions.

2.2 Master Credentials Reference Table

For immediate system demonstration, testing, and operational validation, the following verified credentials are configured across all roles:

Role / Persona	Username	Password	Registered Identifier	Direct Contact
Super Admin	admin	admin123	admin@safepassage.ai	+91 90000 00001
Cab Driver	kumar_driver	driver123	kumar@cabs.com	+91 98401 23456
Cab Owner	ravi_owner	owner123	ravi.owner@chennaicabs.com	+91 98401 55667
Parent	priya_parent	parent123	priya.sharma@gmail.com	+91 98401 11223
Student	mahesh_student	student123	mahesh.k@loyola.edu	+91 98401 99887
Working Pro	vikram_pro	pro123	vikram.m@tcs.com	+91 98401 44332

2.3 Real-Time Approval Status Tracker

Applicants can click '**Check Approval Status**' at any time on the login gateway. Entering their email or phone number displays the real-time review state (Awaiting Super Admin Decision , Active Now , or Accepted for Tomorrow ). Once the Super Admin clicks 'Accept & Activate Now', the account becomes instantly accessible without requiring any secondary code verification.

Section 3: Role 1 — Super Admin Master Console & Central Governance

3.1 Role Purpose & Governance Scope

The **Super Admin Master Console** serves as the central control tower for the entire SafePassage AI network. It provides top-tier oversight across multiple schools, colleges, fleet owners, driver rosters, vehicle compliance databases, financial escrows, and live emergency incidents.

3.2 Core Functional Modules

- 1. Live Fleet Radar & Telemetry Heatmap:** Real-time geographic visualization of all active school buses, college shuttles, and commercial cabs with speed, heading, and delay status.
- 2. Direct Registration Approval & SLA Queue:** Immediate 1-click authorization controls (`■ Accept & Activate Now`, `■ Accept for Tomorrow`, `■ Reject`) with live audit streaming.
- 3. Multi-Tenant Institution Management:** Create, configure, and manage schools and colleges with custom route networks, stop coordinates, and fee structures.
- 4. Financial Clearinghouse & Razorpay Settlements:** Monthly subscription fee audits, platform commission deduction (10–15%), automated driver payouts, and escrow releases.
- 5. Emergency Broadcast & Incident Management:** Immediate one-touch broadcast to all drivers and parents in case of weather emergencies, route blockades, or panic SOS triggers.

3.3 Step-by-Step Super Admin Move Workflow

Step #	Operational Action	System Response & State Transition	Audit & Output
Step 1	Log In as Super Admin	Master password verification (`admin123`). Session initialized with master governance tokens.	Audit log: `Admin session active`
Step 2	Open Approvals Desk	Reviews pending registrations. Cards display requested SLA (`■ Within 30 min` vs `■ Tomorrow`).	Queue displays pending applicant count.
Step 3	Accept Registration	Clicks `■ Accept & Activate Now`. Account state immediately changes to `ACTIVATED`.	Audit log: `APPROVED: User activated`
Step 4	Monitor Live Fleet Radar	Interactive Leaflet map displays real-time GPS locations, route lines, and stop markers.	Active vehicles telemetry refreshed at 3s intervals.
Step 5	Run Financial Settlement	Audits monthly collections, deducts platform fee, dispatches payouts to fleet owners.	Razorpay payout batch receipt generated.

Section 4: Role 2 — Cab Driver Mobile Cockpit & 45-Second Dispatch

4.1 Role Overview & Mobile Experience

The **Cab Driver Mobile Cockpit** (running natively on Android port `8091` or responsive web) empowers school bus drivers, van captains, and on-demand cab operators with distraction-free turn-by-turn navigation, a digital student boarding roster, 45-second fast-dispatch alerts, and instant SOS beacons.

4.2 The 45-Second Fast Dispatch Protocol

When an ad-hoc or single-trip booking is requested, the SafePassage routing engine broadcasts the trip to drivers within a 3.5 km radius. The driver has a **45-second countdown timer** to accept. Upon arrival at the pickup waypoint, passenger boarding is authenticated via a dynamic 4-digit PIN.

45-SECOND SINGLE TRIP PROTOCOL

4-Stage Single Trip Lifecycle:

- 1. **DISPATCH (45s Timer):** Driver receives audible chime and route preview.
- 2. **ACCEPTED & EN ROUTE:** GPS telemetry shared with passenger and parent.
- 3. **ARRIVED & OTP VERIFY:** Driver enters passenger's 4-digit PIN to unlock vehicle doors.
- 4. **TRIP COMMENCED & COMPLETED:** Automated fare settlement and receipt dispatch.

4.3 Digital Conductor Roster & Waypoint Navigation

For scheduled school morning and afternoon routes, the driver cockpit renders an integrated waypoint checklist. As the vehicle approaches each student's home stop (detected via GPS geofencing), the driver marks the student as **Boarded**, **Absent**, or **Dropped Off**. Parents receive automated SMS and in-app push alerts instantly.

Phase	Driver Interaction	Automated System Action
Shift Start	Toggle 'On Duty' in Driver APK.	Vehicle telemetry begins broadcasting to Super Admin & Parent apps.
Route Start	Click 'Begin Morning School Route'.	Route polyline loaded; Next stop ETA calculated and pushed to parents.
Stop Arrival	Vehicle enters 100m geofence radius.	Proximity push alert sent to parent: 'Bus arriving in 2 mins'.
Boarding	Tap student checkbox on digital roster.	Child marked boarded; Timestamp recorded in MySQL database.
School Arrival	Tap 'Complete Morning Trip'.	All students marked safely arrived at school; Trip summary logged.

Section 5: Role 3 — Cab Owner & Fleet Agency Management Portal

5.1 Fleet Operations & Contract Marketplace

Commercial cab owners, school van operators, and corporate fleet contractors utilize the **Cab Owner Portal** to manage multiple vehicles, assign drivers to scheduled routes, monitor fuel efficiency and maintenance schedules, bid on institutional transport contracts, and track net payout revenues.

5.2 Key Fleet Features

- **Vehicle Compliance Vault:** Automated expiry alerts for Fitness Certificates (FC), Commercial Insurance, State Permits, and Pollution Under Control (PUC) certificates.
- **Driver-to-Cab Allocation Matrix:** Dynamic drag-and-drop driver assignment for morning school shifts, afternoon return trips, and evening corporate runs.
- **Contract Bidding Engine:** Bid on exclusive transportation contracts published by partner schools, colleges, and IT tech parks.
- **Fuel & Maintenance Ledger:** Log fuel fills, odometer milestones, tire rotations, and engine servicing with cost-per-kilometer profitability charts.

Fleet Capacity 1 to 50+ Commercial Cabs	Compliance Check Automated RTO Verification	Payout Cycle Weekly / Monthly Automated	Commission Model 10% Platform Fee
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Section 6: Role 4 — Parent Mobile App & Student Safety Portal

6.1 Complete Peace of Mind for Parents

The **Parent Mobile App** (running on port `8092` with dedicated Metro packaging) is engineered to provide absolute transparency into student transit. Parents can track their child's school cab in real time on an interactive radar map, receive proactive proximity alerts, verify driver credentials, mark student absences, and pay monthly transportation fees securely.

6.2 Step-by-Step Parent Daily Move Workflow

Daily Event	Parent Action in App	Live System Feedback
Morning Pickup	Open app to check Live Radar.	Vehicle location, speed (e.g. 34 km/h), and dynamic ETA displayed.
500m Proximity	Receives push notification.	Alert: 'Cab is 500m away. Please prepare child at pickup point.'
Boarding Confirm	App screen updates automatically.	Status changes to: 'Child Safely Boarded at 7:42 AM'.
School Drop-off	Receives arrival confirmation.	Notification: 'Child safely arrived at ABC School campus.'
Leave Marking	Tap 'Mark Absent' if child is sick.	Stop automatically bypassed on driver's morning route; Saves 6 mins.
Fee Payment	Click 'Pay Monthly Fee (Rs. 2,400)'.	Razorpay payment gateway opens; Digital receipt generated instantly.

SAFETY & EMERGENCY PROTOCOL

Direct Driver Emergency Calling: The Parent App features a 1-tap direct call button connecting the parent straight to the assigned cab driver's verified mobile phone without revealing personal private numbers via telephony masking.

Section 7: Role 5 — College Student Commuter Pass & Ride Share

7.1 College Transit & Peer Pooling Network

College and university students need flexible transit options tailored to shifting academic schedules, club activities, and campus peer pooling. The **Student Commuter Hub** provides digital QR passes, live shuttle tracking, and peer ride-sharing with automated split-fare calculations.

7.2 Digital Smart Commute Pass

- **Dynamic QR Pass:** Cryptographically refreshed QR pass scanned by the college bus conductor to record boarding.
- **Live Campus Shuttle Radar:** View all campus-bound buses, current passenger occupancy percentages, and estimated arrival times at designated city stops.
- **Peer Ride Pooling:** Request shared rides with fellow verified students heading in the same direction, splitting fuel costs automatically.
- **Night Exam Shuttle Booking:** Special late-evening bus bookings during semester examinations with campus security escort link.

Section 8: Role 6 — Corporate Working Professional Shuttle Service

8.1 Corporate IT Park & Enterprise Mobility

Working professionals and corporate employees commuting to IT Parks, Special Economic Zones (SEZs), and corporate campuses utilize the **Corporate Shuttle Service**. The platform coordinates desk-to-door dynamic routing, automated expense reimbursement invoices, and late-night shift safety protocols.

8.2 Key Corporate Capabilities

- **Guaranteed Reserved Seating:** Subscribe to monthly morning and evening fixed-time corporate luxury shuttles.
- **Desk-to-Door Route Optimization:** AI clustering calculates the most efficient pickup sequence based on daily employee attendance.
- **GST-Compliant Corporate Invoicing:** Download tax invoices for monthly corporate travel reimbursement.
- **Late-Night Safety Escort:** Automated drop-off confirmation ping required when employee arrives home after 8:00 PM.

Section 9: Native Mobile Applications Ecosystem

9.1 React Native Multi-App Structure

SafePassage AI features three dedicated React Native mobile applications sharing a unified TypeScript design system with optimized build configurations:

Application Name	Directory & Package	Metro Port	Target Device / Emulator
SafePassage Parent App	<code>`frontend-mobile-parent` com.helloworld</code>	8092	Android Phone / Pixel 3a AVD
SafePassage Driver Cockpit	<code>`frontend-mobile-driver` com.helloworld</code>	8091	Android Phone / Pixel 3a AVD
SafePassage Admin Mobile	<code>`frontend-mobile-admin` com.helloworld</code>	8093	Android Phone / Pixel 3a AVD

9.2 Automated Port Reversal & Hardware Acceleration

To guarantee smooth local development and zero-crash execution on Windows machines with Intel UHD Graphics, the automated launcher scripts execute the following pipeline:

- 1. **Emulator Boot:** Launches AVD with ``-no-snapshot -gpu swiftshader_indirect`` to prevent Intel graphics translator crashes.
- 2. **ADB Wait & Boot Detection:** Polls ``sys.boot_completed`` property with 180s cold-boot resilience.
- 3. **Network Port Reversal:** Maps ``adb reverse tcp:8092 tcp:8092`` (Metro) and ``adb reverse tcp:8085 tcp:8085`` (Parent API).
- 4. **Dedicated Metro Window:** Starts independent Metro bundler process on assigned port.
- 5. **Gradle Build & APK Deploy:** Automatically compiles debug APK and launches ``MainActivity`` on screen.

Section 10: Backend Microservices & API Gateway Architecture

10.1 Spring Boot Microservices Decomposition

The backend is architected as six domain-driven Spring Boot 3.2 microservices operating in a decoupled mesh architecture:

Microservice Engine	Port	Core Responsibilities	Key REST Endpoints
Auth & Super Admin Gateway	8080	Master JWT issuing, password management, role registration approval queue, audit streaming.	<code>`/api/auth/login`</code> <code>`/api/admin/requests`</code> <code>`/api/admin/users/status`</code>
School & Institution Service	8081	School/college catalog, student admissions, grade rosters, academic calendar sync.	<code>`/api/schools`</code> <code>`/api/schools/{id}/students`</code>
Fleet & Cab Service	8082	Vehicle compliance docs, maintenance logs, fleet agency profiles, capacity limits.	<code>`/api/fleet/vehicles`</code> <code>`/api/fleet/compliance`</code>
Student & Booking Service	8083	Daily commute bookings, single-trip 45s dispatch requests, Razorpay fee subscriptions.	<code>`/api/bookings/create`</code> <code>`/api/subscriptions/pay`</code>
Driver & Telemetry Service	8084	GPS telemetry ingestion, geofence trigger evaluation, student boarding roster updates.	<code>`/api/driver/telemetry`</code> <code>`/api/driver/roster/board`</code>
Parent & Notification Gateway	8085	Live vehicle radar proxy, proximity push alert dispatcher, driver emergency call link.	<code>`/api/parent/radar/{id}`</code> <code>`/api/parent/leave`</code>

Section 11: Database Architecture & MySQL Relational Schema

11.1 Relational Entity Data Models

Persistence is managed via MySQL 8.0 with InnoDB foreign key constraints, B-Tree indexes on geographical coordinates, and transactional ACID guarantees:

Table Name	Primary Key	Foreign Keys	Key Fields & Indexes
`users`	`id` (BIGINT)	—	`username`, `email`, `password_hash`, `role`, `status`, `phone`
`institutions`	`id` (BIGINT)	—	`name`, `code`, `type` (SCHOOL/COLLEGE), `latitude`, `longitude`, `contact`
`vehicles`	`id` (BIGINT)	`owner_id`	`plate_number`, `vehicle_type`, `capacity`, `fc_expiry`, `insurance_expiry`
`drivers`	`id` (BIGINT)	`user_id`, `vehicle_id`	`license_number`, `badge_number`, `rating`, `duty_status`
`routes`	`id` (BIGINT)	`institution_id`	`route_name`, `start_time`, `end_time`, `polyline_geojson`
`route_stops`	`id` (BIGINT)	`route_id`	`stop_name`, `sequence_order`, `latitude`, `longitude`, `expected_time`
`students`	`id` (BIGINT)	`parent_id`, `institution_id`, `stop_id`	`first_name`, `last_name`, `grade`, `roll_number`, `status`
`telemetry_logs`	`id` (BIGINT)	`vehicle_id`, `driver_id`	`latitude`, `longitude`, `speed_kmh`, `heading`, `recorded_at`
`payments`	`id` (BIGINT)	`user_id`, `student_id`	`razorpay_order_id`, `amount`, `status`, `payment_date`

Section 12: Operational Runbook, Desktop Launchers & Verification Matrix

12.1 1-Click Desktop Launchers Directory

All application components and background services are accessible via preconfigured 1-click desktop batch launchers:

Launcher Shortcut	Target Component	Operational Actions Automated
`Launch_App.bat`	Backends & Web Portal	Starts MySQL check, launches Spring Boot microservices, boots Vite web prototype on `localhost:5173`.
`Run_Parent_App.bat`	Parent Mobile App	Boots Android Emulator with SwiftShader GPU, sets port reverse (8092 & 8085), starts Metro, installs debug APK.
`Run_Driver_App.bat`	Driver Mobile App	Boots Android Emulator, sets port reverse (8091 & 8084), starts driver Metro bundler, installs driver APK.
`Start_Android_Emulator.bat`	Standalone Emulator	1-Click standalone GUI launcher for `Pixel_3a_API_34_extension_level_7_x86_64` virtual device.

12.2 System Verification & Compliance Sign-Off

Subsystem / Capability	Validation Method	Verification Result
Unified Role-Based Gateway	Manual credential entry tested across all 6 personas.	PASS ■ (Strict Auth Active)
Direct Super Admin Approval	■ 30-min urgent & ■ tomorrow batch workflows tested.	PASS ■ (Instant Sign-in)
Android Emulator Automation	SwiftShader GPU boots without Intel UHD crashes.	PASS ■ (Pixel 3a Online)
React Native Parent APK	Build debug APK on port 8092, verified screen capture.	PASS ■ (Render Verified)
React Native Driver APK	Build debug APK on port 8091, 45s dispatch ready.	PASS ■ (Cockpit Verified)
Web Prototype Build	TypeScript compile + Vite build (1826 modules).	PASS ■ (0 Errors)
Spring Boot Microservices	Maven wrapper clean build on Java 17.	PASS ■ (All 6 Services)
MySQL Relational Schema	DDL schema integrity and foreign key constraints.	PASS ■ (Production Ready)

FINAL ARCHITECTURAL COMPLIANCE & APPROVAL

Architectural Sign-off: The SafePassage AI platform has completed end-to-end integration across all frontends, mobile APKs, microservices, and databases. This master brochure document serves as the official operational reference and system workflow specification.